

Basic Principles of Medicine 1
Module : Cardio, Pulmonary, Renal
Lecture: CPR 34

Mechanics of Lung & Chest

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Learning Objectives

- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0204** Explain surface tension and how it applies to lung mechanics, including the effects of alveolar size and the role of surfactants. Infer the role of surfactants in preventing atelectasis.
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0205** Illustrate and explain the pressure-volume (compliance) curves for the lungs, chest wall, and respiratory system on the same set of axes. Show and explain the significance of the resting positions for each of these three structures.
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0206** Identify the forces that generate the negative intrapleural pressure when the lung is at functional residual capacity and interpret the direction that the lung and chest wall will move if air is introduced into the pleural cavity (pneumothorax).
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0207** Explain how lung and chest wall compliance interact to set FRC and how changes in lung or chest wall recoil can change in a patient with obstructive/restrictive disease or with pneumothorax.

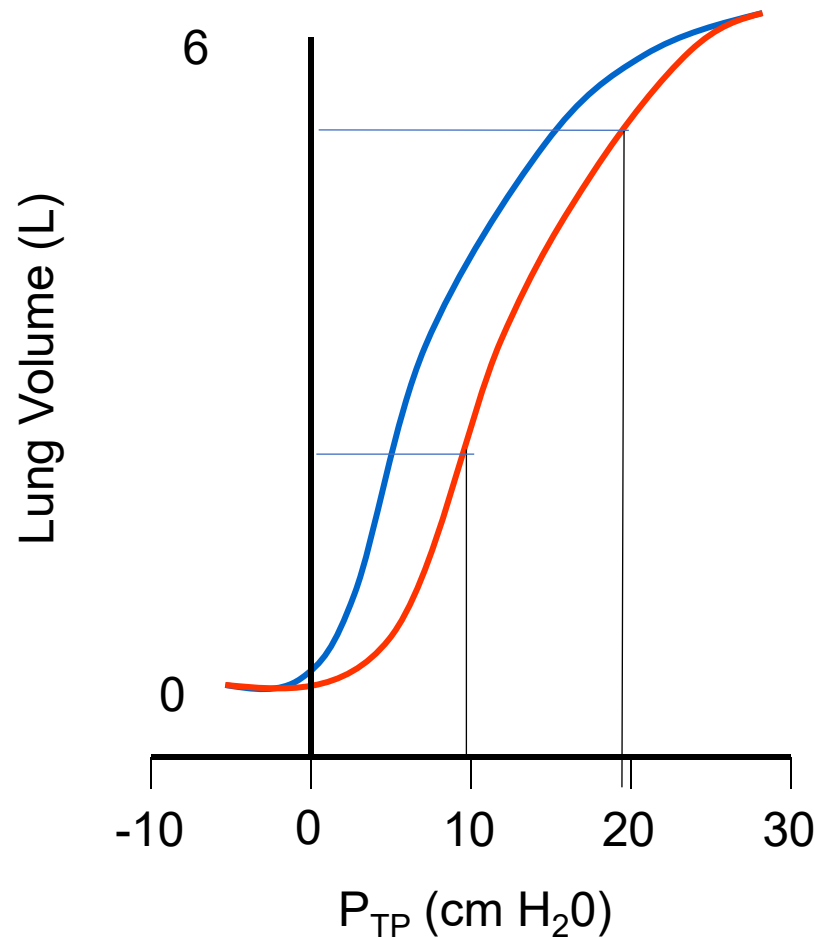
Recommended Reading

Rhoades and Bell, Medical Physiology Principles for clinical medicine, 6e, Chapter 18, Ventilation and the Mechanics of Breathing.

Online Link: <https://meded-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=253088352&bookid=3211>

Slides with blue background are the supplemental information of the previous slides and thus, might not be explained again during the lecture time. However, please read the slides for clarification as they are equally important to know

Lung Compliance (C_L): Pressure Volume Relationship

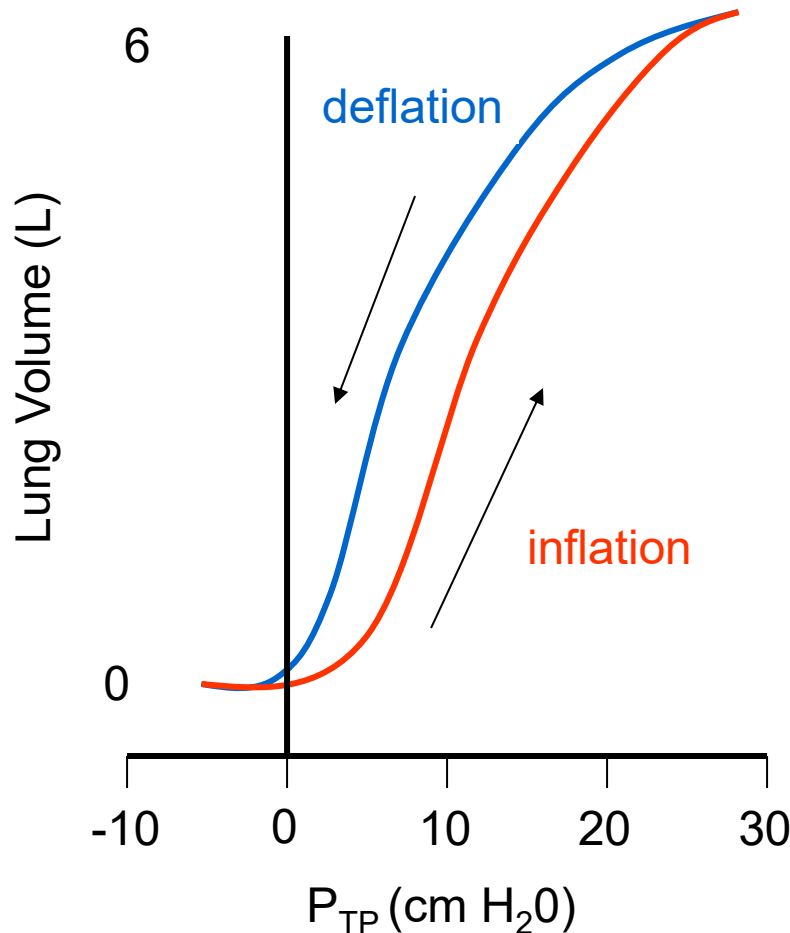


$$C_L = \frac{\Delta V_L}{\Delta P_{TP}}$$

$$C_L = \frac{1}{E}$$

E = Elastance

Lung Compliance: Pressure Volume Relationship



Hysteresis:

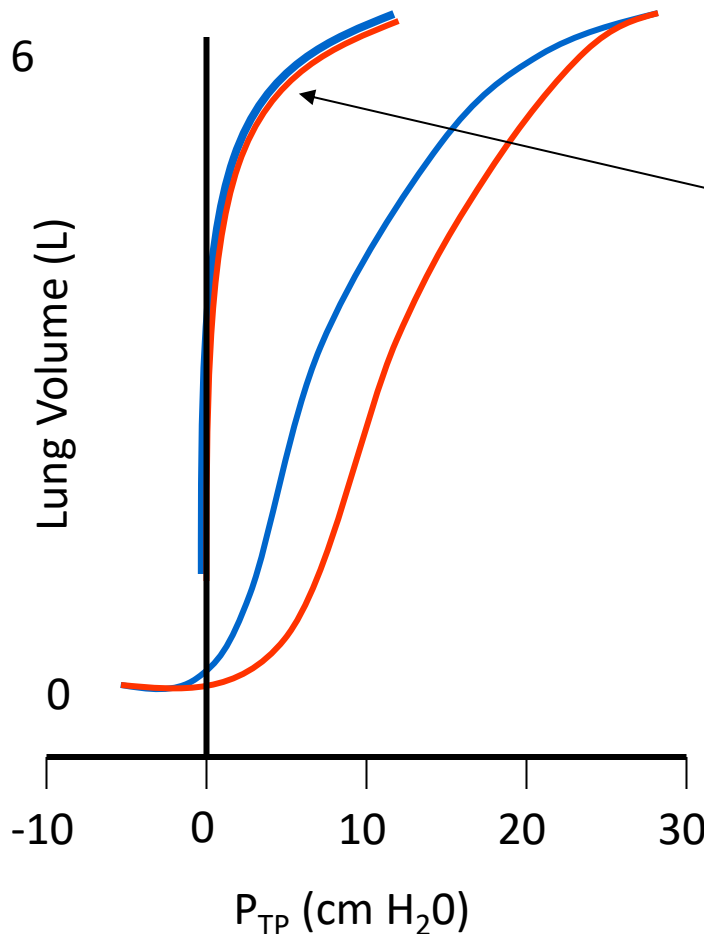
The pressure/volume relationship differs depending on inflation or deflation

Reasons for Hysteresis:

1. Surface Tension
2. Elasticity of lung

It is easier to deflate lungs than to inflate

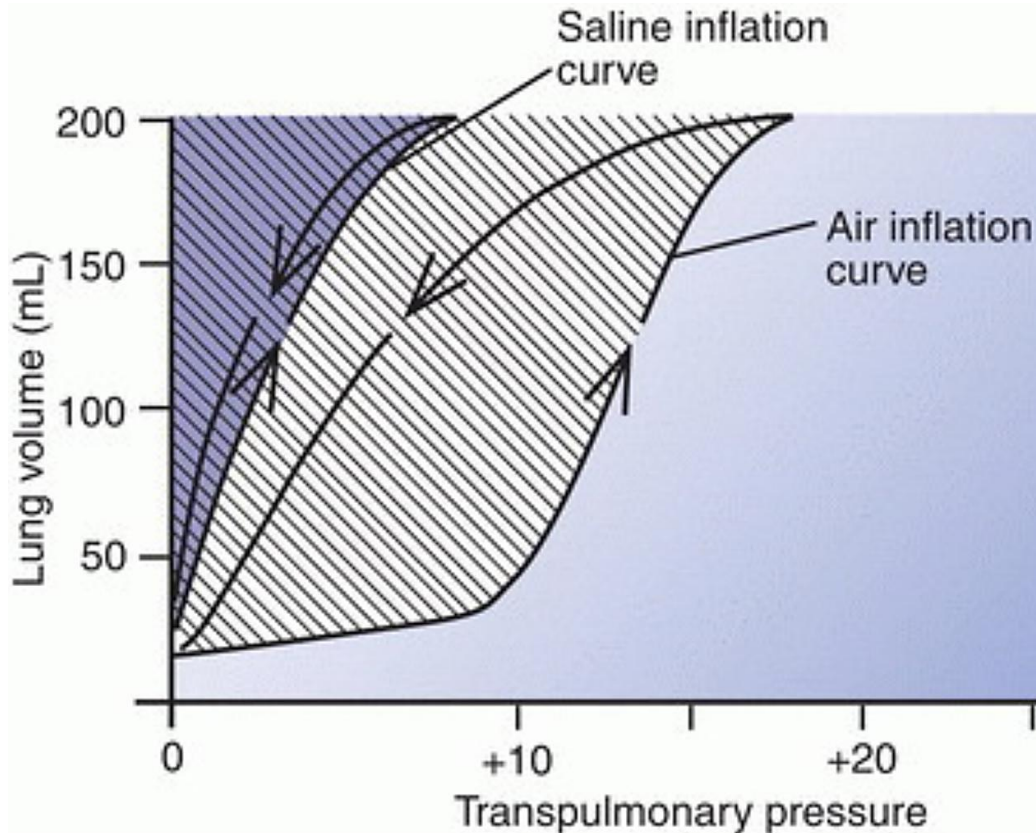
Lung Compliance: Pressure Volume Relationship



If lung is filled with saline to eliminate surface tension, hysteresis is not seen

Major factor for hysteresis during inflation-deflation is: **Surface Tension** force in the alveoli.

Surface Tension and lung volume affect Lung Compliance (C_L)

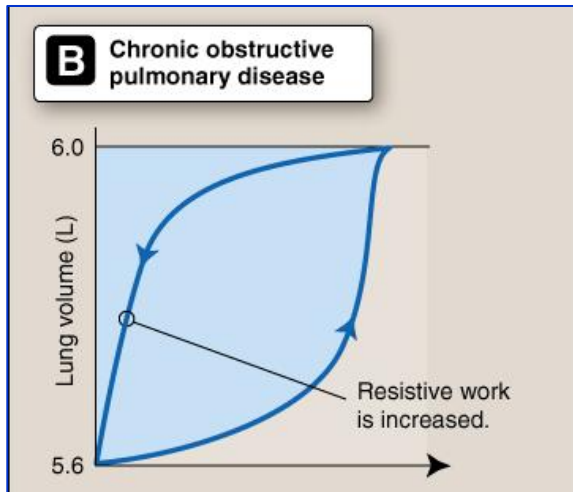


- Also, note that in air inflated lung, the lung compliance is low at both high and low lung volumes.
- Lung compliance is high (steep part of the curve) at moderate lung volumes.

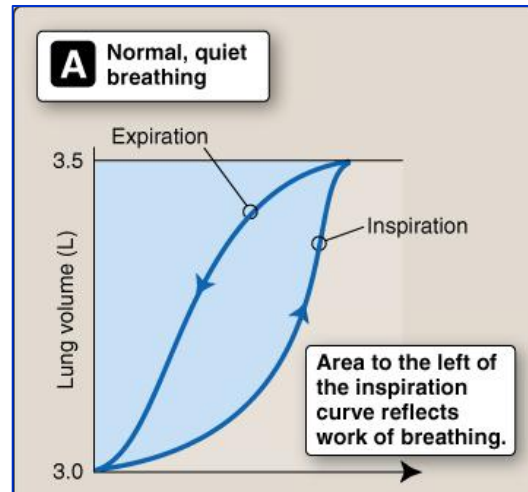
Explanation for the figure in the previous slide:

Alveolar **surface tension** affects lung compliance. Inflating lungs with saline instead of air eliminates surface tension. Two pressure–volume curves for lungs inflated to the same volume are shown, first with air and then with saline. The differences in the two curves occur because surface tension contributes significantly to lung compliance in the air-filled lungs. When lungs are inflated and deflated with saline, the lungs are more distensible, with a concomitant increase in lung compliance. The shaded areas are equal to the work done in inflating the lungs.

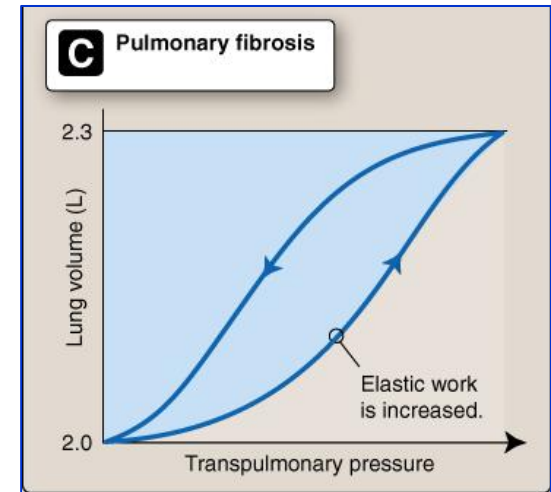
Effects of pulmonary disease on the work of breathing



Patient 1



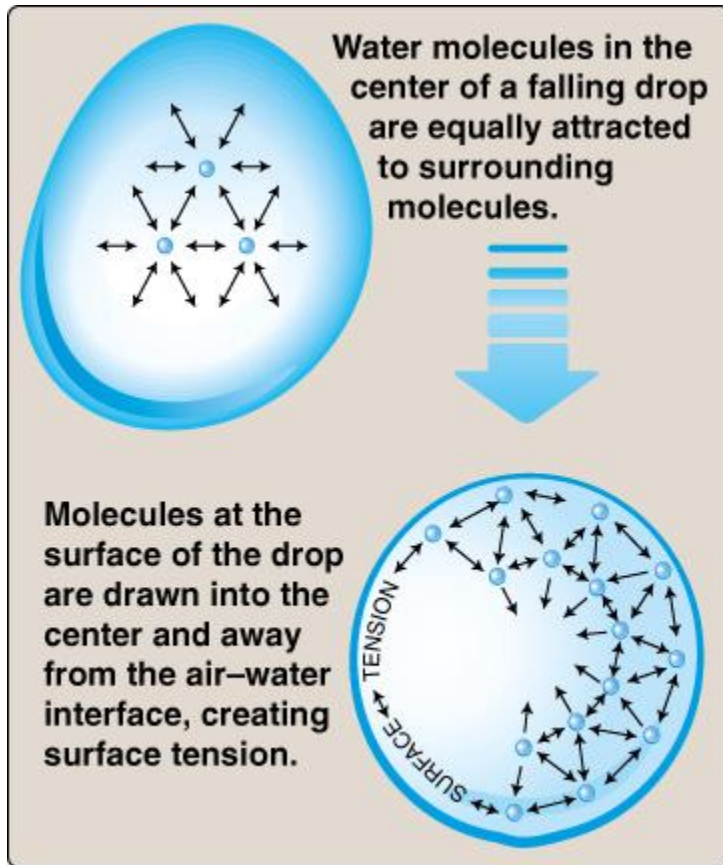
Normal
Healthy
Subject



Patient 2

Surface Tension

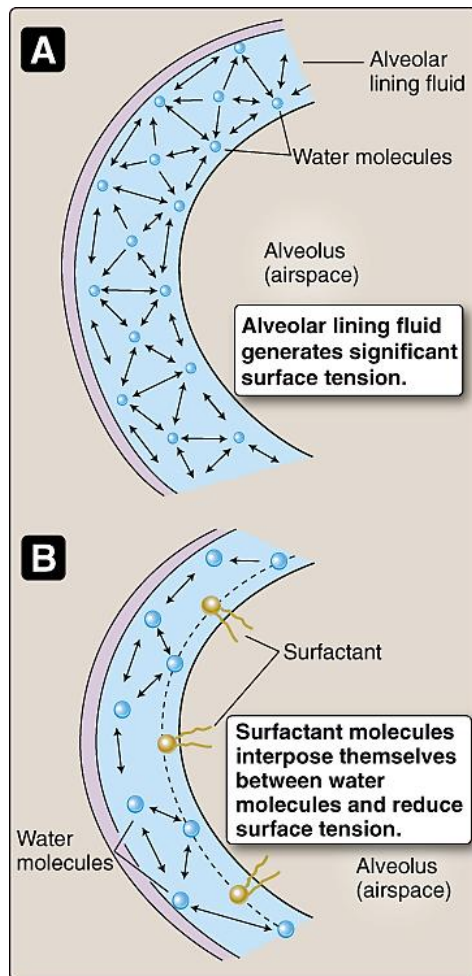
Origins of surface tension.



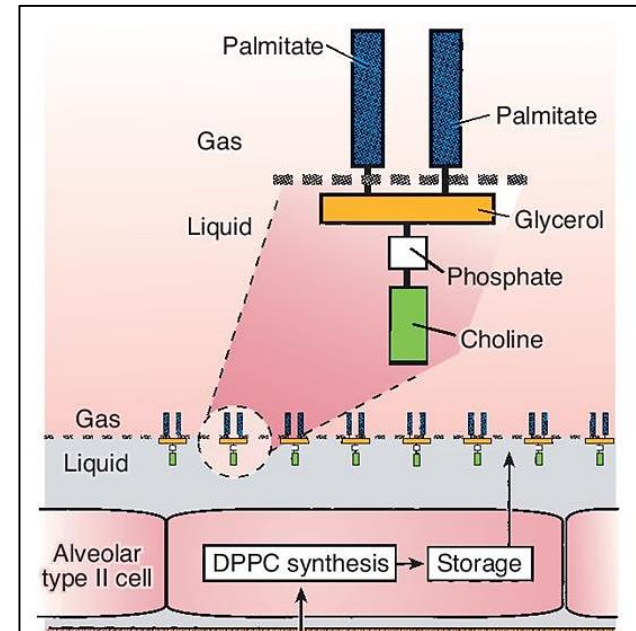
- Intermolecular forces of attraction – in bulk, a molecule is exposed to these forces from all around
- Property of the surface of a liquid that allows it to act as an elastic sheet
- Force per unit length/energy per unit area
- Molecules at surface are only attracted from below and with each other
- As a result, the surface of a liquid acts as an elastic sheet
- Measured as Force per unit length or energy per unit area
- Smallest surface area for a given volume is a sphere

Surface Tension lowering capability of Lung surfactants

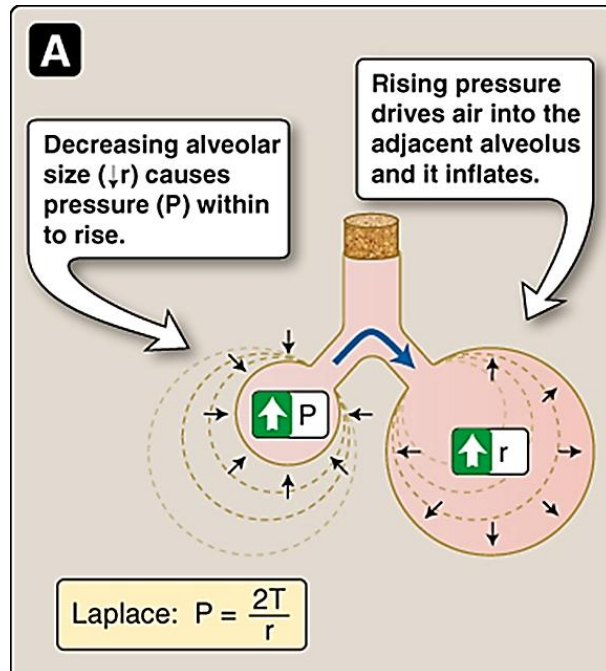
**Higher the concentration,
lower the surface tension**



**Surfactant effects on surface tension
created by alveolar lining fluid.**



Surface Tension and Alveolar instability (Role of Laplace's law)



$\uparrow$ radius $\downarrow$ P
 $\downarrow$ radius $\uparrow$ P

Small alveoli develop a large pressure
Large alveoli develop a smaller pressure

Creates a pressure gradient between
alveoli of different sizes

Air from the smaller tends to flow into
the larger alveoli

Smaller alveoli tend to collapse, and
larger alveoli tend to overinflate

Creates instability in the alveoli during
breathing

Lung Surfactants prevent this to happen How?

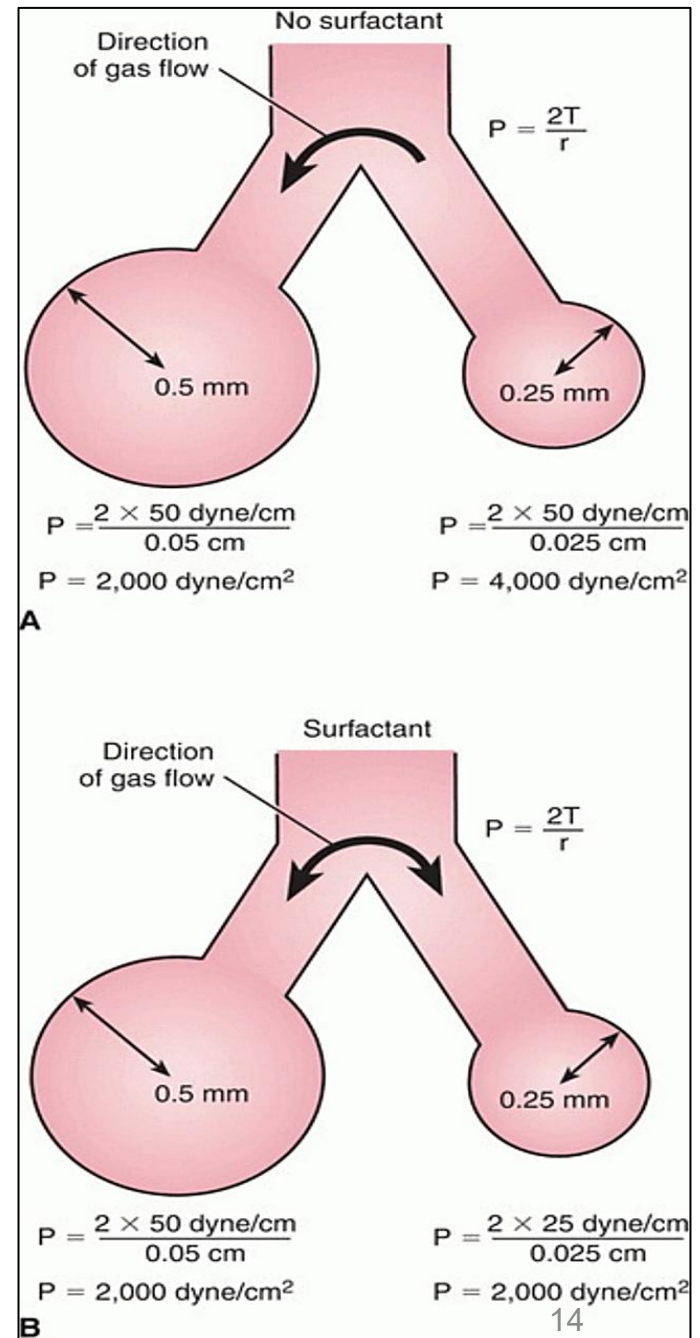
Surfactant promotes alveolar stability

(A) If surface tension remains constant (50 dyne/cm), alveoli that are interconnected but differ in diameter become unstable and cannot coexist. Pressure in the smaller alveolus is greater than that in the larger alveolus, which causes air from the smaller alveolus to empty into the larger alveolus. At low lung volumes, the smaller alveoli tend to collapse, a phenomenon known as

atelectasis.

(B) **Surfactant lowers surface tension proportionately more in the smaller alveolus.** As a result, pressures in the two alveoli are equal, and alveoli of different diameters can coexist.

P, pressure; r, radius; T, tension.



Surfactant Constituents

Component	%	Name and Function
Phospholipids	90	Mostly Dipalmitoyl phosphatidylcholine (DPPC) and Some Phosphatidylglycerol. Function: Reduce surface tension and improve lung compliance.
Protein	10	Surfactant proteins A, B, C and D. Function: Facilitate surfactant monolayer formation; Play role in lung defense.

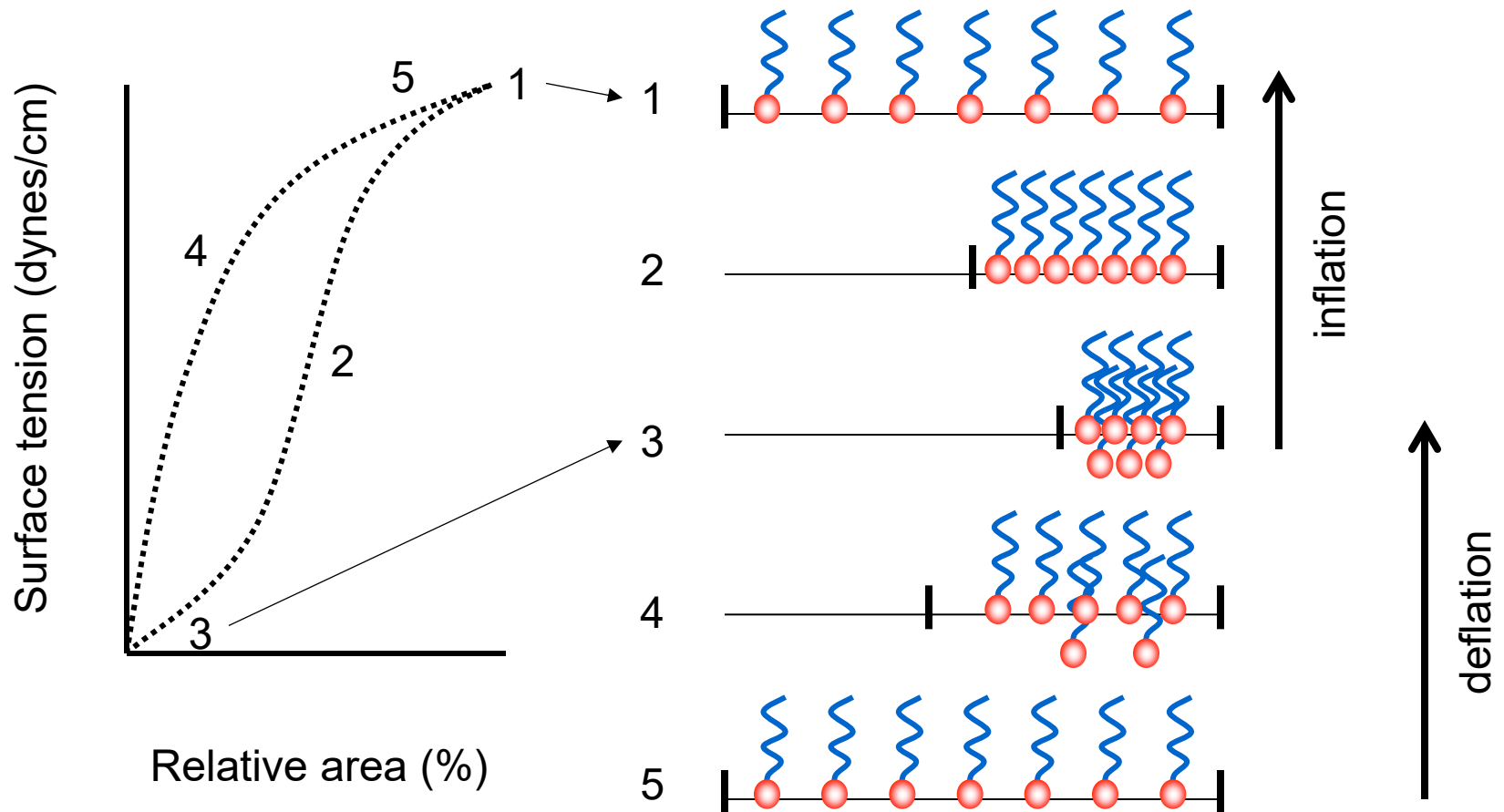
**Babies born prematurely
have low surfactants &
may develop Neonatal
Resp Distress Syndrome**



Lung Surfactant

- **Surface Active Agent**
- Pulmonary surfactant is a lipoprotein rich in phospholipid
- Secreted by type II cells in the alveolus
- Hydrophobic and hydrophilic regions
- Main component **dipalmitoyl phosphatidylcholine (DPPC)**
- Reduces surface tension greatly (reduces tendency to collapse) and thus **increases lung compliance**
- Surface tension lowering ability of the lung surfactants is **concentration (surface density) dependent**. Lower the size of the alveoli, surfactant concentration increases, and surface tension falls to a greater extent.
- Lung surfactants **stabilize alveoli**.
- Lung surfactants reduce the tendency of fluid movement from pulmonary capillaries into the interstitial space and thus, keep **alveoli dry**.

Surfactant Distribution in the alveoli during Respiratory cycle

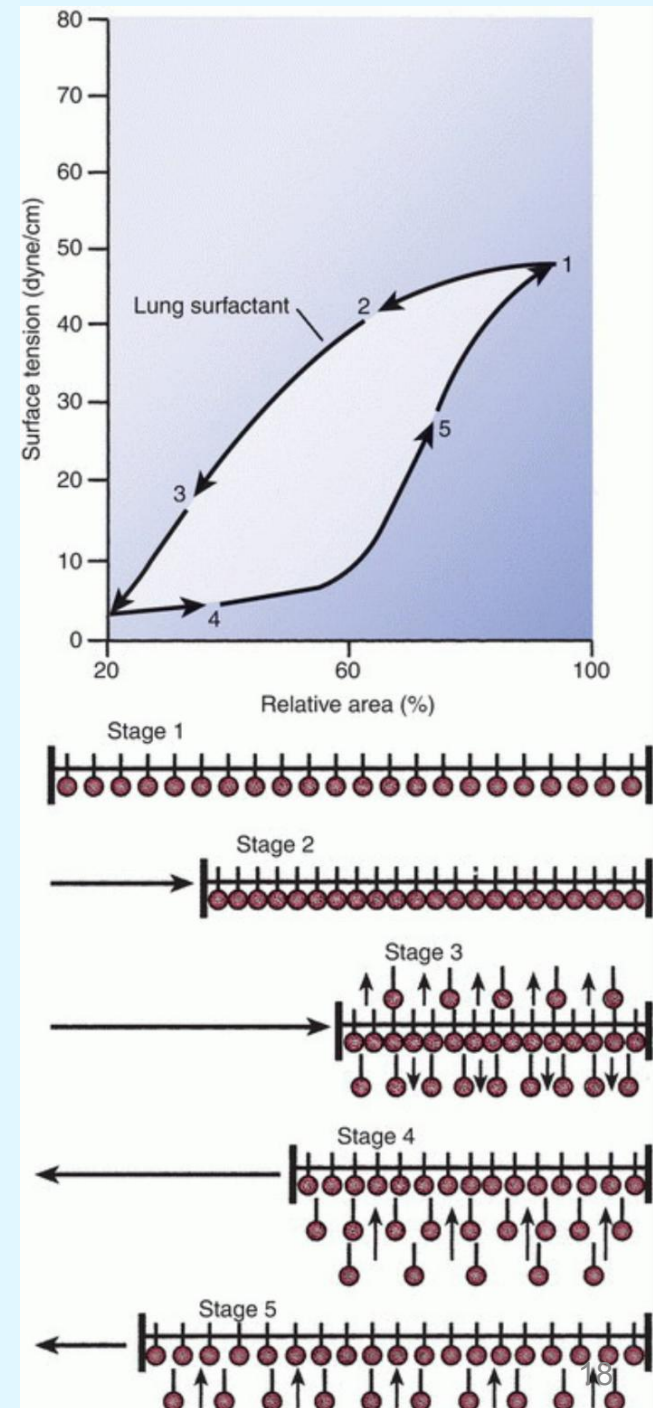


Biophysical mechanism of lung surfactant for lowering alveolar surface tension.

Surfactant molecules are compressed during lung deflation. At stage 3, surfactant molecules form micelles and are removed from the surface.

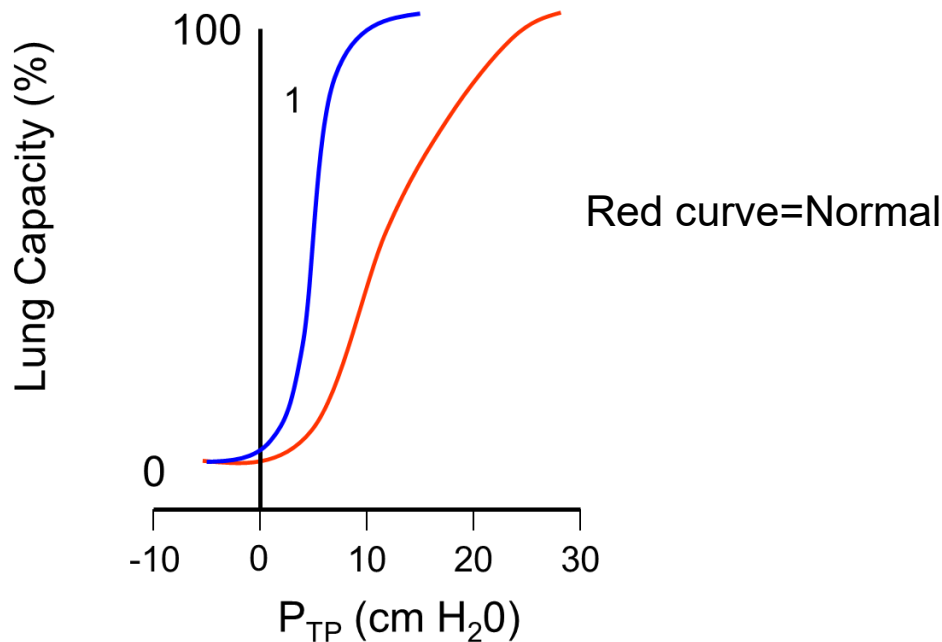
On lung inflation, new surfactant is spread onto the surface film (stage 4).

Turnover of lung surfactant is high because of continual replacement of surfactant during lung expansion.



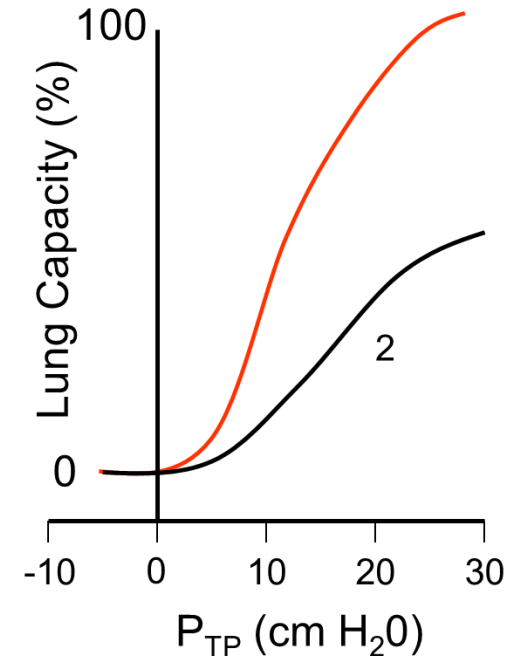
Static Compliance (C) changes in Disease States

High compliance



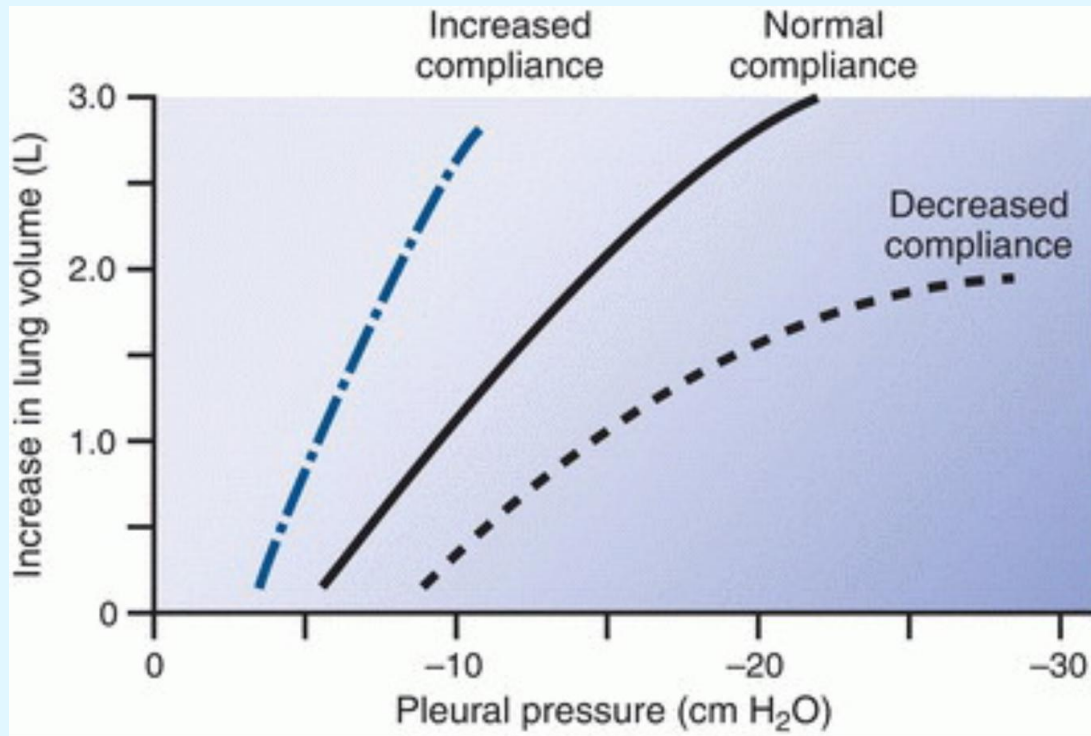
- Lung inflates easily and has little elastic recoil
- Example: Emphysema/ COPD

Low compliance



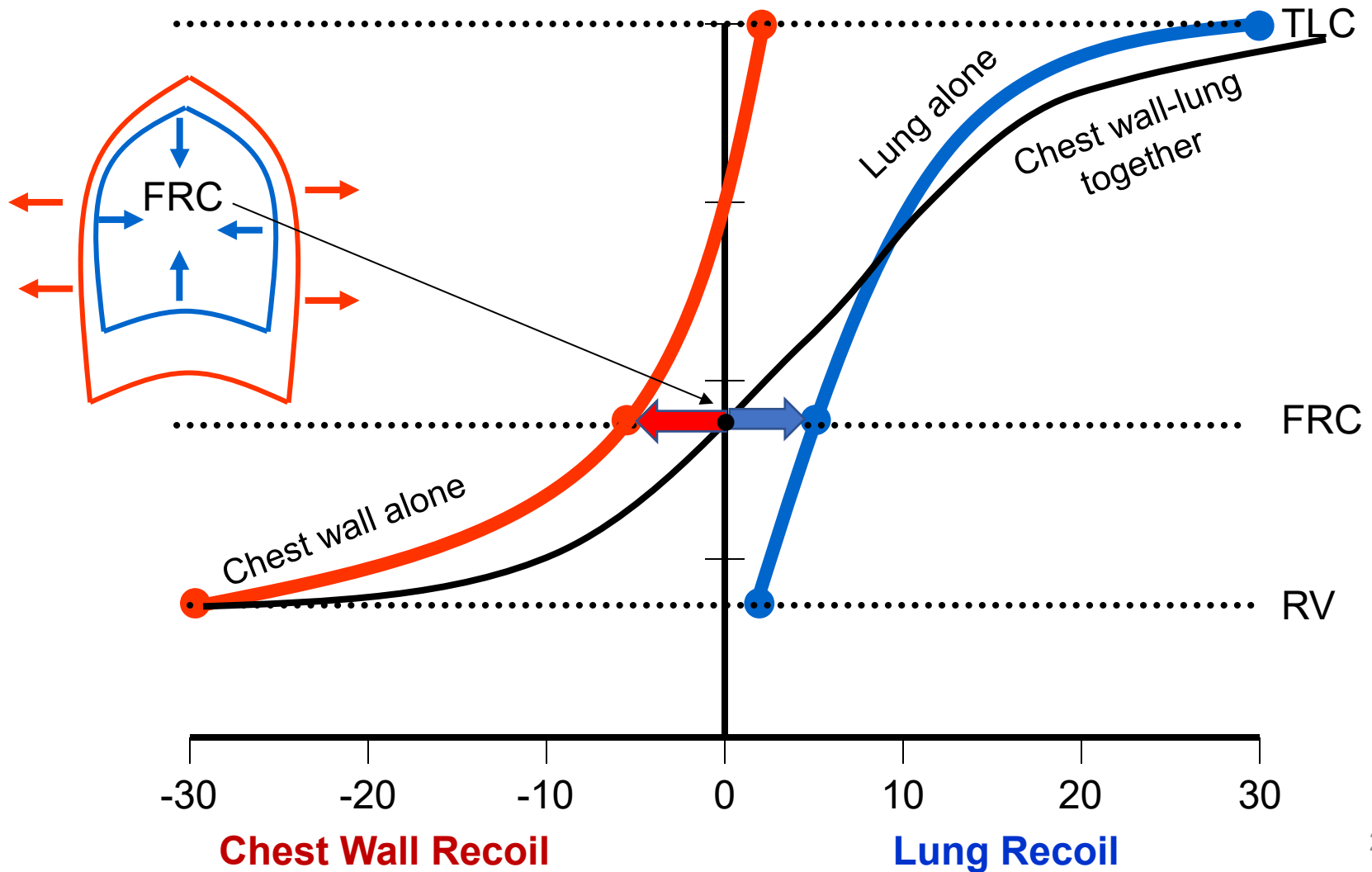
- Lung inflates with difficulty due to large elastic recoil
- Example: Pulmonary fibrosis, Respiratory distress syndrome

DIY task: How would you categorize two different patients with the different static compliance curves as shown in the following image?

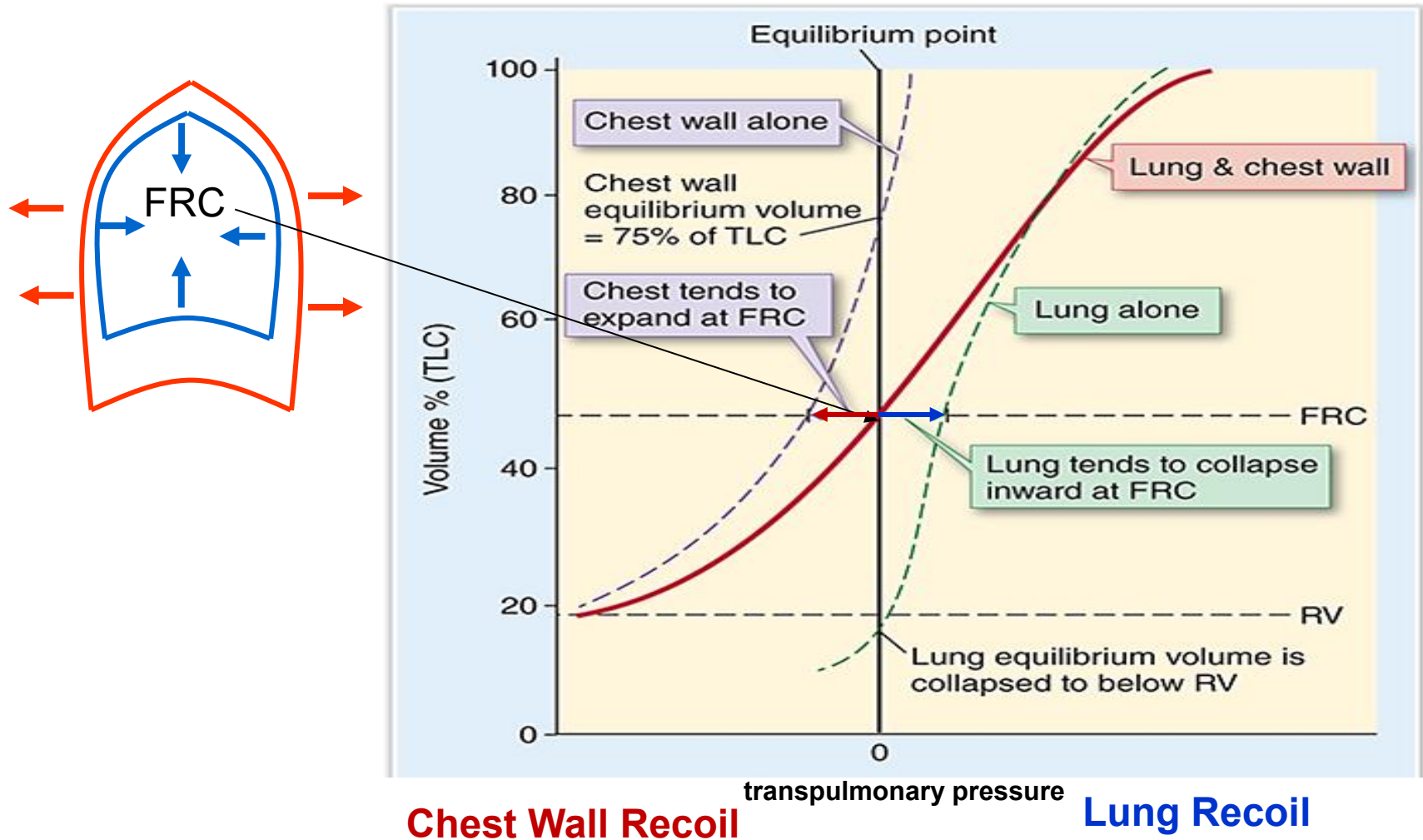


Obstructive and restrictive diseases alter lung compliance. Patients with a chronic obstructive lung disease, such as emphysema, have abnormally high lung compliance. Patients with restrictive diseases, such as respiratory distress syndrome or pulmonary fibrosis, have abnormally low lung compliance.

Chest Wall and Lung Recoil Pressures Together



Chest Wall and Lung Recoil Pressures



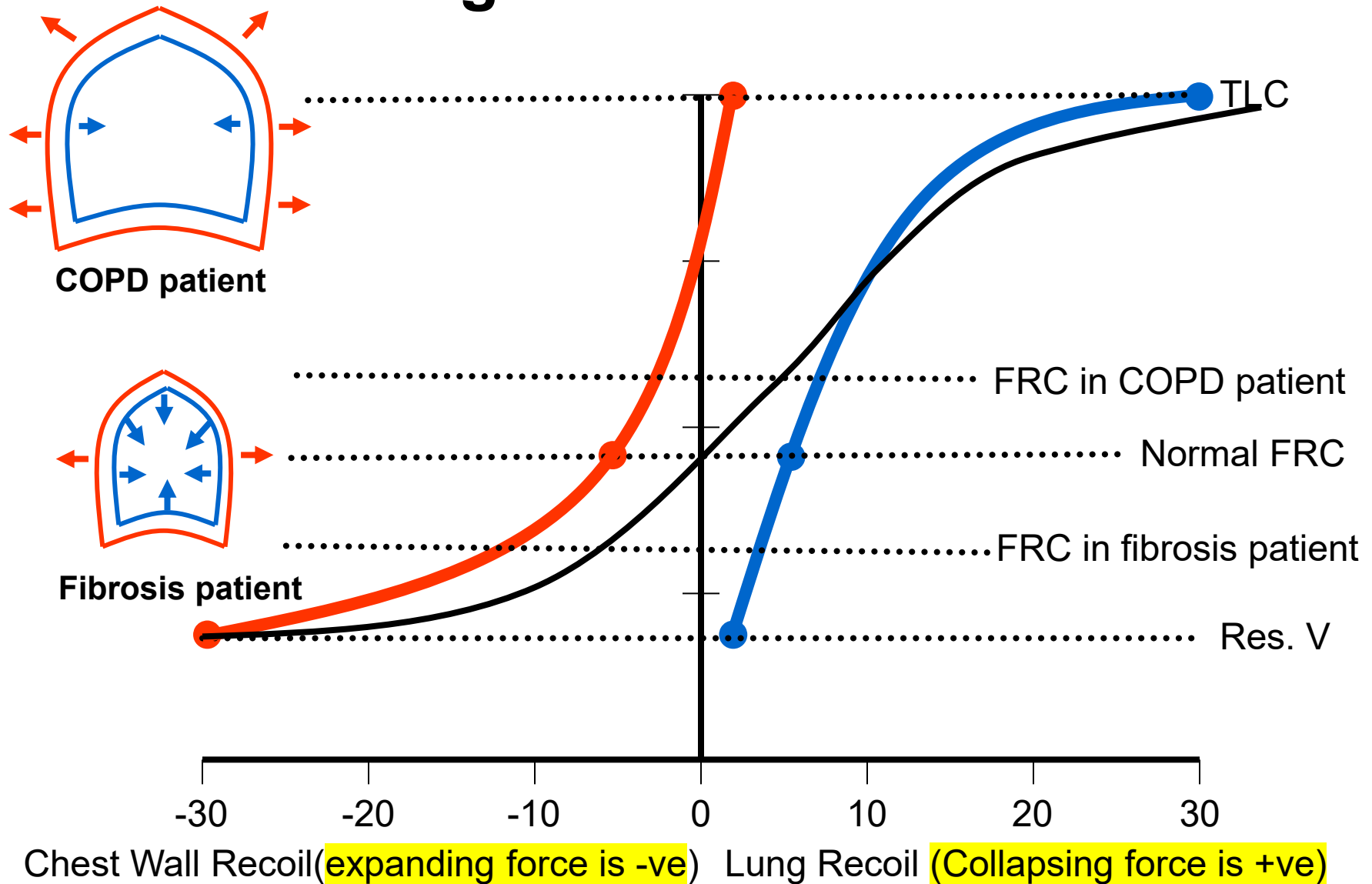
Functional Residual Capacity (FRC)

- Functional Residual Capacity
- Volume of the lung at rest between breaths
- No net recoil because this is the point where the outward recoil of the chest wall is counterbalanced by the inward recoil of the lungs.
- The whole respiratory system is in its relaxed state.

Passive Recoil Forces set at FRC

- FRC is the equilibrium volume at which the inwardly directed lung recoil is **exactly balanced** by the outwardly directed chest wall recoil (spring out).
- The sum of these 2 passive recoil forces is zero at FRC.
- FRC is the resting volume of the lung when there is no active muscle contraction.
- Sets ventilatory tone.

Chest Wall and Lung Recoil Pressures Together in diseases



Chest Wall and Lung Recoil Pressures Together in diseases: Explanation slide

Why FRC is high in COPD patients?

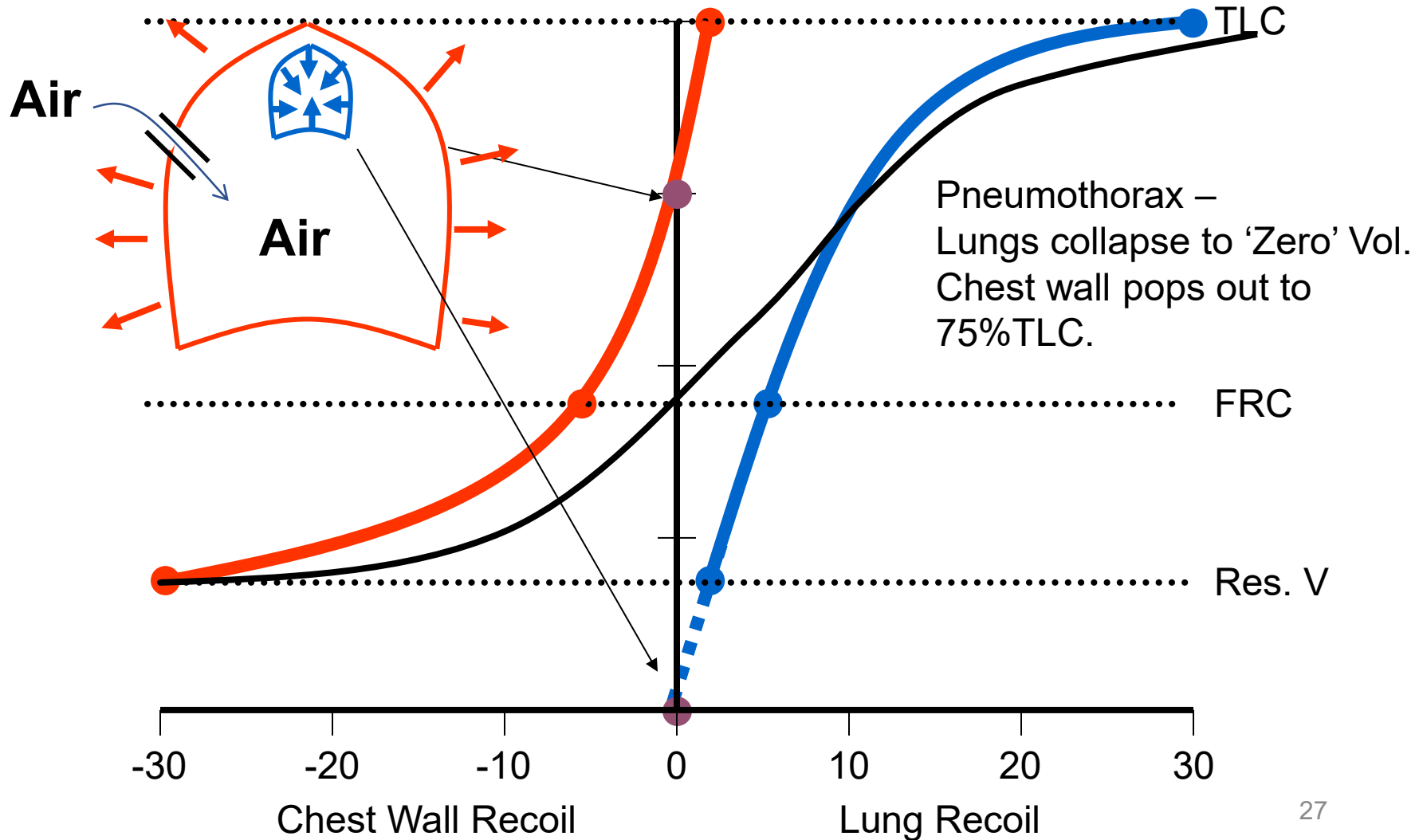
In a patient with obstructive lung disease like emphysema, an increase in lung compliance makes the lung compliance curve **steeper**. As a result, at a given volume, the collapsing (elastic recoil) force on the lungs is decreased. At the original value for FRC, the tendency of the lungs to collapse is less than the tendency of the chest wall to expand, and these opposing forces will no longer be balanced. For the opposing forces to be balanced, *extra volume must be added to the lungs to increase their collapsing force*. Thus, the combined lung and chest-wall system seeks a **new higher FRC**. With this higher FRC, the two opposing forces can be balanced and the new intersection point, where airway pressure is zero, is at a **higher FRC**. Therefore, a patient with emphysema breathes at higher lung volumes with a higher FRC. These patients have a **barrel-shaped chest**.

Chest Wall and Lung Recoil Pressures Together in diseases: Explanation slide

Why FRC is low in patients with restrictive lung disease?

Pulmonary fibrosis is a restrictive pulmonary disease in which lungs are stiff (decreased compliance). A decrease in lung compliance makes a **decreased slope** of the lung compliance curve. At the original FRC, the tendency of the lungs to collapse is greater than the tendency of the chest wall to expand and the opposing forces will no longer be balanced. To reestablish balance, the lung and chest-wall system will seek a new FRC which is **lower than the normal FRC**. The combined lung and chest-wall system finds a new intersection point, where airway pressure is zero, at a **lower FRC**. Therefore, a patient with fibrosis will have a lower FRC (shrunk lung). Because of lower FRC, these patients also have **low residual volume (RV) and total lung capacity (TLC)**.

Chest Wall and Lung Recoil Pressures Together in pneumothorax



Dynamic Compliance: Factors influencing it

- Definition: The change in lung volume during cyclic breathing is called dynamic lung compliance (C_{dynamic}).
- Static and dynamic lung compliances are the same at zero respiratory rate.
- In a normal healthy person, the dynamic compliance decreases slightly (approx. 5%) if the respiratory rate increases from 0 to 48/min.
- If the airway resistance increases, C_{dynamic} falls significantly as respiratory rate rises from 0 to 48 breaths/min.
- Importance: In patients with increased airway resistance (asthma patients), the dynamic compliance decreases significantly with increased respiratory frequency.

Any Questions ?

Please reach me:

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